

David Feldman

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EDUCATION

University of Illinois Urbana-Champaign

Urbana, IL

B.S. Mathematics and Computer Science - GPA: 4.0

Expected Grad May 2028

- State Farm CS Scholarship: 1 of 2 from a cohort of 2,500+ recognized for exceptional academic promise
- Societies: James Scholar Honors Program, Quant @ Illinois (Software Division), SIAM, ACM SIGma, SIGPLAN
- Coursework: Computational Geometry, Algorithms and Models of Computation, Computer Architecture, Data Structures

RESEARCH

Illinois Mathematics Lab

Jan 2026 — May 2026

UIUC Math Department

Urbana, IL

- Researched computational approaches to homotopy groups of spheres, fundamental in higher-dimensional topology
- Built optimized Rust and Python tools that computed further into the EHP spectral sequence than the algorithm's author
- Produced the most detailed visualizations of the sequence to date; presented findings in a research poster ([repo](#))

SELECTED PROJECTS

Real-Time Audio Visualization Suite

Nov 2025

- Manual Rust implementation of iterative Cooley-Tukey FFT; real-time spectrogram with synchronized WAV playback
- Lock-free SPSC ring buffer for cross-thread audio streaming; memory-mapped WAV parser with zero-copy sample reinterpretation and lazy bidirectional sliding-window cache

High Performance Particle Physics Engine

Mar 2025

- Multithreaded physics engine simulating kinematics, gravity, and collisions for 50,000+ particles at 60 FPS in Rust
- Lock-free parallelization with atomics, grid spatial partitioning and sweep for collision detection; Verlet integration

Eulerian Fluid Simulation

Aug 2025

- Interactive real-time 2D CFD simulator with incompressible flow, obstacle interaction, and multiple visualization modes
- MAC grid velocity field with backward advection sampling, divergence-free projection for pressure solving in Rust

Constraint Satisfaction Puzzle Solver

Jul 2025

- [Flow Free](#) game clone and CSP solver with constraint propagation and forward checking heuristics in Rust
- Automated puzzle collection pipeline via Python web scraper with 1500+ extracted levels

Automatic Differentiation Engine

Jul 2025

- Rust clone of Karpathy's micrograd; automatic differentiation engine, topological dag sort for arbitrary expression gradient
- Used to implement backpropagation fundamentals; usable as library dependency for neural network training

End-to-end Mechanical Keyboard

Dec 2023

- Fully custom printed PCB with [QMK/C](#), community group buy and open-sourced case hardware and firmware

SKILLS

- **Languages:** Rust, Python, C/C++, OCaml, Typst, LaTeX
- **Tools:** Git, Cargo, rayon, cargo-flamegraph, PyTorch, NumPy
- **Concepts:** Numerical Methods, Differentiable Programming, Performance Optimization, Computational Geometry, Linear Algebra, Algebraic Topology, Concurrency

EXPERIENCE

Full-Stack Developer

Sep 2023 — May 2025

Tech Team, Los Altos Hacks

Los Altos, CA

- Developed software infrastructure for the world's largest high school hackathon for two years in a row
- Coordinated feature development on multi-team React/TypeScript projects through Git workflows
- Led the judging system sub-team, implementing custom pairwise voting to streamline in-hackathon voting